# OOP using C++

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# C++

C++ is a general-purpose programming language

Implementations of C++ exist from some of the most modest microcomputers to the largest supercomputers and for almost all operating systems.

C++ is a superset of the C programming language. In addition to the facilities provided by C, C++ provides flexible and efficient facilities for defining new types.

The C++ language support four programming styles:

- Procedural programming
- Data abstraction
- Object-oriented programming
- Generic programming

C++ supports systems programming.

**History:**

C++ developed by **Bjarne Stroustrup** in 1979 at AT&T Bell Laboratory.

C++ developed using  class concept of Simula( already existing language which supported OOP paradigm) and C's  efficiency and flexibility for systems programming.

Templates and exceptions added to C++ later.

**1979:** Work on ''C with Classes'' started. The initial feature set included classes and derived classes, public/private access control, constructors and destructors, and function declarations with argument checking.

**1984:** ''C with Classes'' was renamed to C++. By then, C++ had acquired virtual functions, function and operator overloading, references, and the I/O stream, number libraries.

**1985:** First commercial release of C++

**1991:** The C++ Programming Language, Second Edition, presenting generic programming using templates and error handling based on exceptions

**1997:** The C++ Programming Language, Third Edition [Stroustrup,1997] introduced ISO C++, including namespaces, dynamic_cast, and many refinements of templates. The standard library added the STL framework of generic containers and algorithms

**2011:** ISO C++11 standard was formally approved. In 2012 The first complete C++11 implementations emerged. 2012 Work on future ISO C++ standards (referred to as C++14 and C++17) started

**2017:** the final standard was published in December 2017.[

**Need of C++**

C++ is a systems programming language used for developing native applications

Software written in C++ is everywhere:  It is in your computer, your phone, your car, and other many electronic devices

Many operating systems have been written in C++ like Windows, Apple's OS, Linux, and most portable-device OSs..

C++ is used to write device drivers and other software that rely on direct manipulation of hardware under real-time
constraints.

C++ is used to write some critical parts of most widely used systems like Amazon, Google, Facebook.

Many other advanced technologies depends on C++'s performance and reliability in their implementations –

JVM of Java Technologies, Web services framework of Microsoft's .NET technologies, JavaScript interpreters of many browsers like Microsoft's Internet Explorer, Mozilla's Firefox, Apple's Safari, and Google's Chrome.

Games has been another major applications area for C++.

Massive use of C++ in demanding embedded systems, projects for flight control software, automobile software, engineering computation is done in C++.

**First C++ Program :**

```
#include<iostream>
using namespace std;
int main()
{
   cout<<"Hello World..";
   return 0;
}
```

**Tokens in C++** : In C++, tokens can be defined as the smallest building block of C++ programs that the compiler understands. Every word in a C++ source code can be considered a token.

Before translation parser does parsing of each statement and finds out tokens. Each token has some pre-defined meaning in compiler. Compiler translates each token in machine understandable form and returns .exe file. This .exe file is executable file, which can be executed directly on OS prompt.

Types of tokens :

- Identifiers
- Keywords
- Constants
- Datatypes
- Strings
- Operators
- Escape sequence

**Identifier :** It is user-defined word. Used to define and declare variable_name, function_name, user_defined data types etc..

Rules to define identifier:

1. It should start with a alphabet or _ (underscore)
2. White space and special characters are allowed in name of identifier
3. It can have alphabet, digits and _ (underscore)
4. It cannot be a keyword
5. It must be unique (can only be declared once in a namespace)
6. C++ is case sensitive, so be careful.

**Keywords : C++ is case sensitive. All keywords are in lower-case.**

| asm | auto | bool | break | case | catch | char |
|---|---|---|---|---|---|---|
| class | const | const_cast | continue | default | delete | do |
| double | dynamic_cast | else | enum | explicit | export | extern |
| false | float | for | friend | goto | if | inline |
| int | long | mutable | namesapce | new | operator | private |
| protected | public | register | reinterprete _cast | return | short | signed |
| sizeof | static | static_cast | struct | switch | template | this |
| throw | true | try | typedef | typeid | typename | union |
| unsigned | using | virtual | void | volatile | wchar_t | while |

#define MAX 100

Preprocessor directive, works like find and replace. Before compilation all occurrences of word MAX in program are replaced by 100.

**Constants :**

const int i = 10;
The value of i will remain unchanged for entire program.

constexpr double area = 3.14 * r * r;
The expression is evaluated at compile time and value is kept in area name, which is constant.

**Demonstration : ConstDemo.cpp**

**Datatypes:**

We use the variables to store values in programs, but the OS has to know what kind of data we want to store in them, since OS is going to allocate some amount of memory to store a value.

A declaration is statement that introduces a variable name into the program. It specifies a datatype for the variable

A datatype defines a set of possible values and a set of operations we can perform on those values.

C++ offers a variety of primitive (built-in) datatypes.

**Datatypes supported in C++ :** Size and range depends on the system the program is compiled for,

| Name | Description | Size* | Range* |
|---|---|---|---|
| char | Character or small integer. | 1byte signed: | -128 to 127 unsigned: 0 to 255 |
| short int (short) | Short Integer. | 2bytes | signed: -32768 to 32767<br>unsigned: 0 to 65535 |
| int | Integer. | 4bytes | signed: -2147483648 to 2147483647<br>unsigned: 0 to 4294967295 |
| long int (long) | Long integer. | 4bytes | signed: -2147483648 to 2147483647<br>unsigned: 0 to 4294967295 |
| bool | Boolean value. | 1byte | true or false |
| float | Floating point number. | 4bytes | +/- 3.4e +/- 38 (~7 digits) |
| double | Double precision | 8bytes | +/- 1.7e +/- 308 (~15 digits) |
| long double | Long double precision | 8bytes | +/- 1.7e +/- 308 (~15 digits) |
| wchar_t | Wide character. | 2 or 4 bytes | 1 wide character |

**Demonstration : Limits.cpp**

**string in C++** :

Variables that can store non-numerical values that are longer than one single character are known as strings.

The C++ language library provides support for strings through the standard string class. This is not a primitive type, but it behaves like a primitive types in its most basic usage.

**Demonstration : stringDemo.cpp**

**Operator:** C++ has very rich set of operators.

**Assignment operator** =

The part at the left of the assignment operator (=) is known as the lvalue (left value) and the right one as the rvalue (right value).
The lvalue has to be a variable whereas the rvalue can be either a constant, a variable, the result of an operation or any combination of these.
The most important rule when assigning is the right-to-left rule:
The assignment operation always takes place from right to left, and never the other way:

```
a = b;
```

This statement assigns to variable a (the lvalue) the value contained in variable b (the rvalue).
The value that was stored until this moment in a is not considered at all in this operation, and in fact that value is lost.

A property that C++ has is that the assignment operation can be used as the rvalue (or part of an rvalue) for another assignment operation.
For example: a = 2 + (b = 5);

is equivalent to:

b = 5; a = 2 + b;

that means: first assign 5 to variable b and then assign to a the value 2 plus the result of the previous assignment of b (i.e. 5), leaving a with a final value of 7.

The following expression is also valid in C++:

a = b = c = 5;  It assigns 5 to the all the three variables: a, b and c

**Arithmetic operators** ( +, -, *, /, % ) : Need 2 operands.

The five arithmetical operations supported by the C++ language are:

+ addition
- subtraction
* multiplication
/ division
% modulo

**Compound assignment** (+=, -=, *=, /=, %=, >>=, <<=, &=, ^=, |=)

**Increment and decrement** (++, --)

**Relational and equality operators** ( ==, !=, >, >=, <, <= )

**Logical operators** ( !, &&, || )

**Comma operator** ( , )

**Explicit type casting operator**

**dot operator (.)**

**Arrow operator ( -> )**

**Conditional operator or Ternary operator** ( ? : )

**Bitwise Operators ( &, |, ^, ~, <> )**

**sizeof() operator**

**Scope resolution operator (::)**

**Demonstration : operatorDemo.cpp**

## Special Characters and Escape Sequence :

| Sequence | Meaning |
|---|---|
| \a | Alert. |
| \b | Backspace. |
| \f | Form feed. |
| \n | Newline. |
| \r | Carriage return. |
| \t | Horizontal tab. |
| \v | Vertical tab. |
| \\ | Backslash ( \ ). |
| \′ | Single quote ( ' ). |
| \" | Double quote ( " ). |
| \? | Question mark ( ? ). |

**Demo: escSequence.cpp**